

Lecture 18

Gram-Schmidt Process & Determinant I

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Roadmap of Semester

Segment 1 Solving linear systems (Lec 3-15);

Segment 2 **Lec 16-22:** Three relatively independent parts:
—**Orthogonality.** Least squares; orthogonal matrix (Lect 16,17,18)
—**Determinant.** [Important tool!] Lec 18,19
—**Linear transformation.** Lec 20-22 [Advanced math perspective of matrix]

They are not directly related to solving system/problem,
but are **fundamental and useful tools**.

Segment 3 **Lec 23-28: Eigenvalues and singular values.**

Today's Lecture: Outline and Learning Goals

Main topics:

1. Gram-Schmidt process
2. Motivation of determinant: Condition for Invertibility

Strang's book: Sec 5

After the lecture, you should be able to

1. Compute orthonormal basis by Gram-Schmidt process
2. Tell the geometrical meaning of determinant

Review

Recall: Equivalent Conditions for Invertibility

Theorem (Equivalent Conditions for Invertibility)

Let $A \in \mathbb{R}^{n \times n}$

The following statements are equivalent:

1. A is **invertible**
2. The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution $\mathbf{x} = \mathbf{0}$
3. A is a product of elementary matrices
4. A has n pivots; or equivalently: $\text{rank}(A) = n$
5. The columns of A span $\mathbb{R}^n$
6. The columns of A are linearly independent
7. The columns of A form a basis of $\mathbb{R}^n$
8. $\dim(C(A)) = n$
9. $\dim(N(A)) = 0$ or $N(A) = \{\mathbf{0}\}$

Orthonormal Basis (标准正交基)

Definition 17.3

A set of vectors $S = \{v_1, \dots, v_n\}$ is called an **orthonormal basis** of a linear space V if

- (1) S is an orthonormal set
- (2) Vectors in S form a basis of V .

Theorem 17.1 (Representation via Orthonormal Basis)

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be an **orthonormal basis** of $\mathbb{R}^n$.

Any vector v can be represented as a linear combination

$$v = \sum_{i=1}^n u_i u_i^\top v = \sum_{i=1}^n \langle v, u_i \rangle u_i$$

Theorem 17.2 (Projection when Ortho-Basis Exists)

Let W be a subspace of $\mathbb{R}^n$ and suppose $S = \{w_1, \dots, w_k\}$ is an **orthonormal basis** of W .

The projection of v onto W can be represented as

$$p = \sum_{i=1}^k \langle w_i, v \rangle w_i$$

Part 0 Exercise: Computing Projection onto Line

How to Compute Projection: Example of Line

Definition 16.1 [Lec 16]

Suppose S is a subspace of $\mathbb{R}^m$.

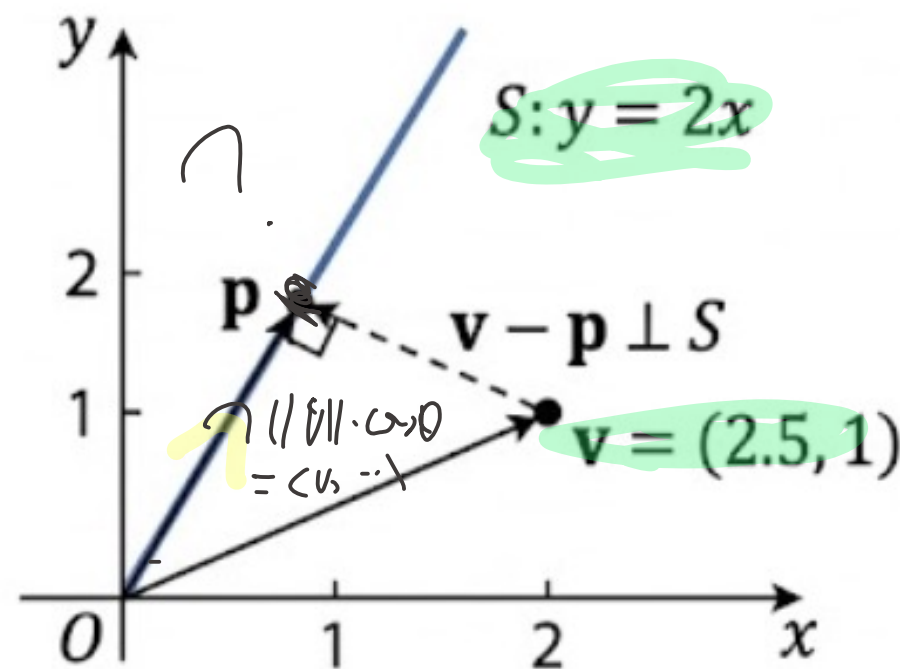
Suppose $\mathbf{p} \in S$ and $\mathbf{b} - \mathbf{p} \perp S$, then we say $\mathbf{p}$ is the projection of $\mathbf{b}$ onto S .

Notation: Denote p as $\text{Proj}_S(b)$

How to compute a projection?

Example:

How to compute p
in this figure?



How to Compute Projection to a Line?

Problem: compute the projection of $\vec{b}$ onto $\text{span}\{\vec{u}\}$: denote as $\vec{p}$.

1st Method: Solving via Orthogonality

Suppose $\vec{p} = \alpha \vec{u}$. *What's α ?*

The error vector $(\vec{b} - \vec{p})$ must be orthogonal to $\vec{u}$, leading to the following derivation:

$$\vec{u}^T (\vec{b} - \vec{p}) = 0 \implies \vec{u}^T (\vec{b} - \alpha \vec{u}) = 0 \implies \vec{u}^T \vec{b} - \alpha \vec{u}^T \vec{u} = 0$$

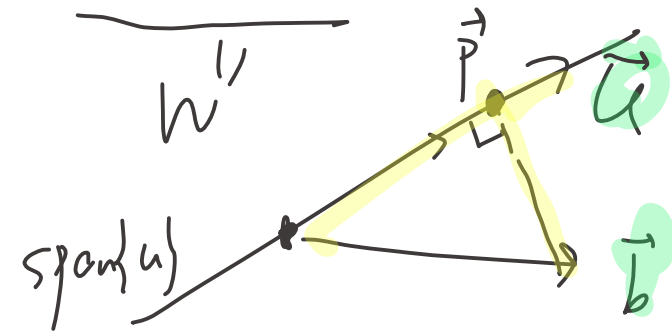
$$\implies \alpha = \frac{\vec{u}^T \vec{b}}{\vec{u}^T \vec{u}} = \frac{\vec{u}^T \vec{b}}{\|\vec{u}\|^2} \implies \vec{p} = \alpha \vec{u} = \left(\frac{\vec{u}^T \vec{b}}{\|\vec{u}\|^2} \right) \vec{u}$$

2nd Method: Use Thm 17.2

$\{\vec{w}_i\}$ basis of W . $\text{Proj}_W(\vec{b}) = \sum_{i=1}^k \langle \vec{b}, \vec{w}_i \rangle \vec{w}_i$

The orthonormal basis for $\text{span}\{\vec{u}\}$ is $\vec{v} = \frac{\vec{u}}{\|\vec{u}\|}$.

By Theorem 17.2, the projection is $\vec{p} = (\vec{b}^T \vec{v}) \vec{v} = \left(\frac{\vec{u}}{\|\vec{u}\|} \right)^T \vec{b} \frac{\vec{u}}{\|\vec{u}\|} = \frac{\vec{u}^T \vec{b}}{\|\vec{u}\|^2} \vec{u}$.



Part I Gram-Schmidt Process

Key Questions in this Part

Thm 17.2: A formula to compute the projection of b onto W , given that the basis of W is known.

For a given subspace V , does an orthonormal basis exist
& how to compute an orthonormal basis?

e.g. $Z = \begin{bmatrix} 2 & 4 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$ Does $C(Z)$ have an orthonormal basis?

We will show: for any subspace S of $\mathbb{R}^n$:

- i) **Existence:** it has an orthonormal basis;
- ii) **Algorithm:** starting from its basis $\{u_1, \dots, u_k\}$, there is a way to find an orthonormal basis of S .

Part I.1 Derivation: $k=2,3$

Is Finding Orthonormal Basis Simple?

Sometimes, yes. e.g. $\mathbb{R}^n$; e.g. column space of

$$\begin{array}{ccccc} & l_1 & l_2 & & l_3 \\ \begin{bmatrix} 1 & 0 & a & 0 & c \\ 0 & 1 & b & 0 & d \\ 0 & 0 & 0 & 1 & e \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Sometimes, no. e.g. Column space of $Z = \begin{bmatrix} -a & -c \\ -b & -d \\ 1 & 0 \\ 0 & -e \\ 0 & 1 \end{bmatrix}$

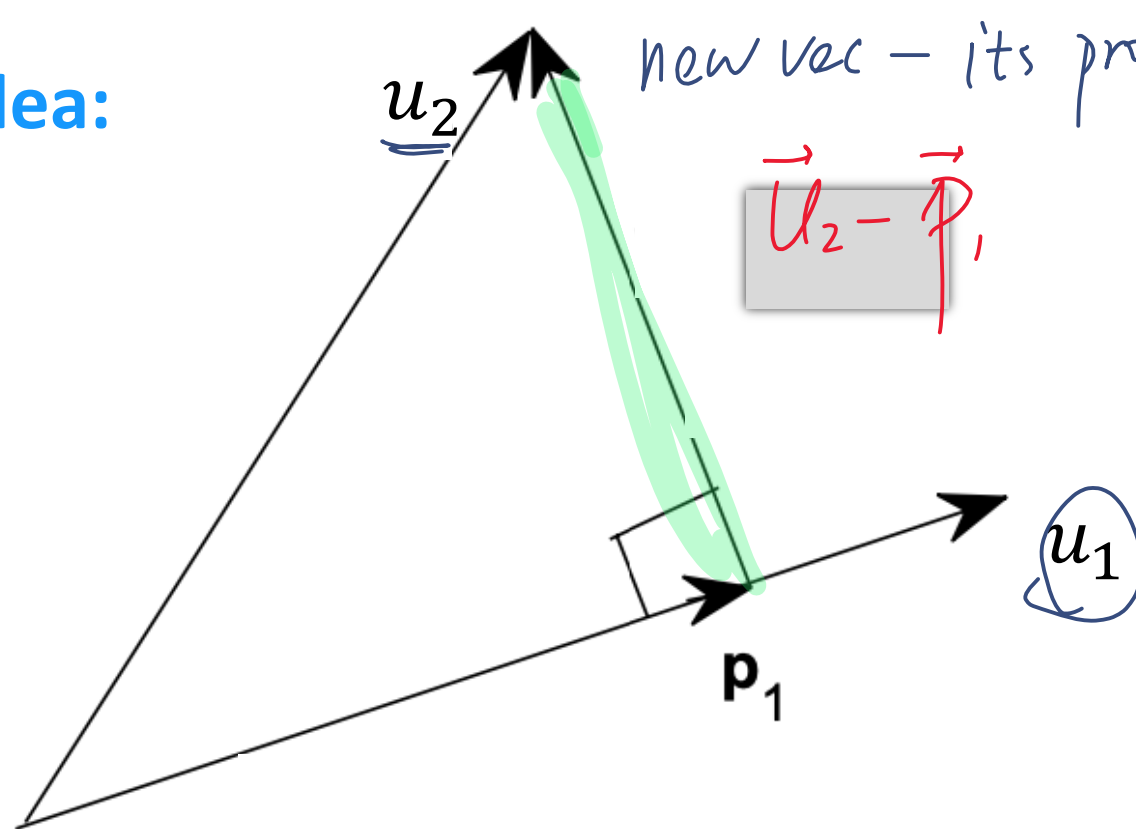
Idea: Use linear combination of **columns** to find orthonormal basis of a column space.

2-Vector Case

$$Z = \begin{bmatrix} 2 & 4 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} = [u_1, u_2]$$

Exercise: Find an orthonormal basis of $\text{span}(\{u_1, u_2\})$

Idea:



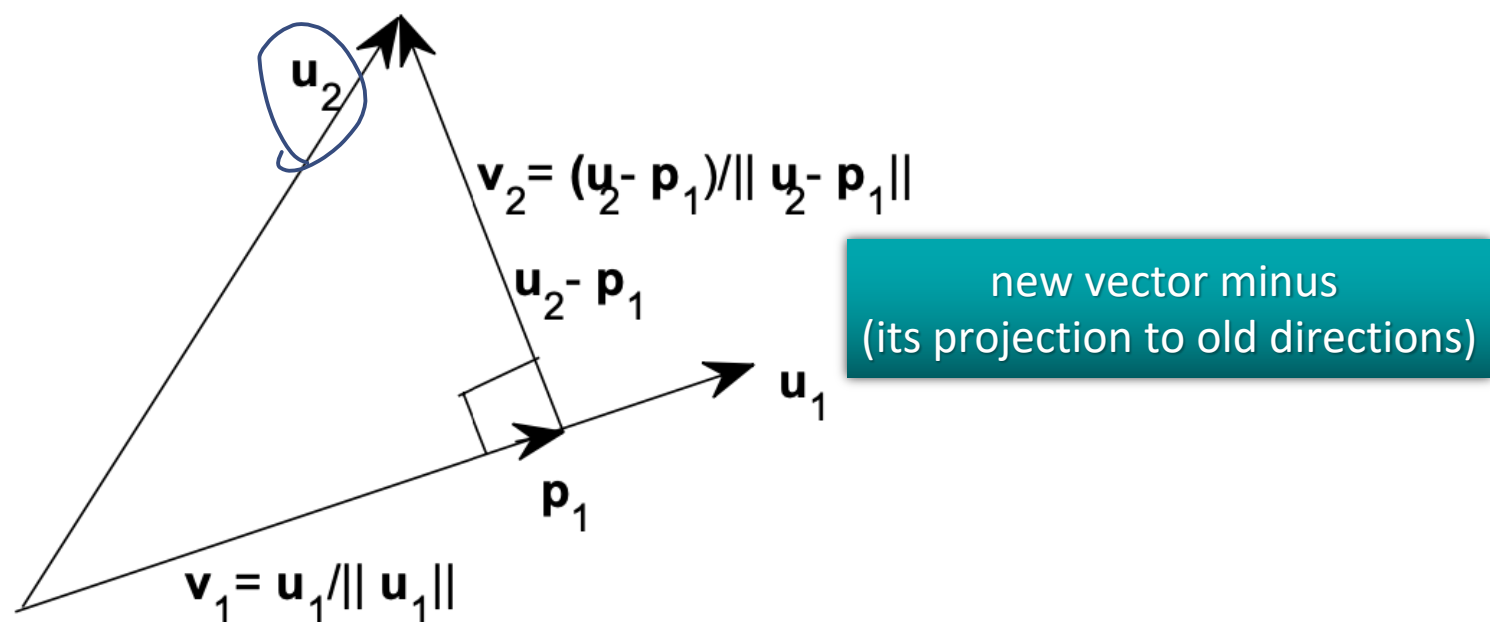
$$v_1 = \frac{u_1}{\|u_1\|}$$

$$v_2 = ?$$

Exercise: Simple Case

$$Z = \begin{bmatrix} 2 & 4 \\ 1 & 0 \\ 0 & 0 \end{bmatrix} = [u_1, u_2].$$

Exercise: Find an orthonormal basis of $\text{span}(\{u_1, u_2\})$



$$v_1 = \frac{u_1}{\|u_1\|} \quad \|v_1\| = 1$$

$$v_2 = \frac{u_2 - p_1}{\|u_2 - p_1\|} \quad \|v_2\| = 1$$

$$\text{where } p_1 = (u_2^T v_1) v_1$$

Coefficient determined by Thm 17.2.

Design Principles of k=2 Case

Goal: Find orthonormal set V s.t. $\text{span}(V) = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

Idea: Induction (“数学归纳”).

Step 1: Find the orthonormal basis $\{v_1\}$ of $\text{span}\{u_1\}$.

Step 2: Find the second vector $\mathbf{v}_2$ in the ortho-basis of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

Design principle: This vector should be

new vector minus
its projection to old directions

i) Orthogonal to $\mathbf{u}_1$;
ii) in $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

Answer: Direction same as $u_2 - p_1$
Where $p_1 = \text{Proj}_{\text{span}\{u_1\}}(u_2) = \langle u_2, v_1 \rangle v_1$

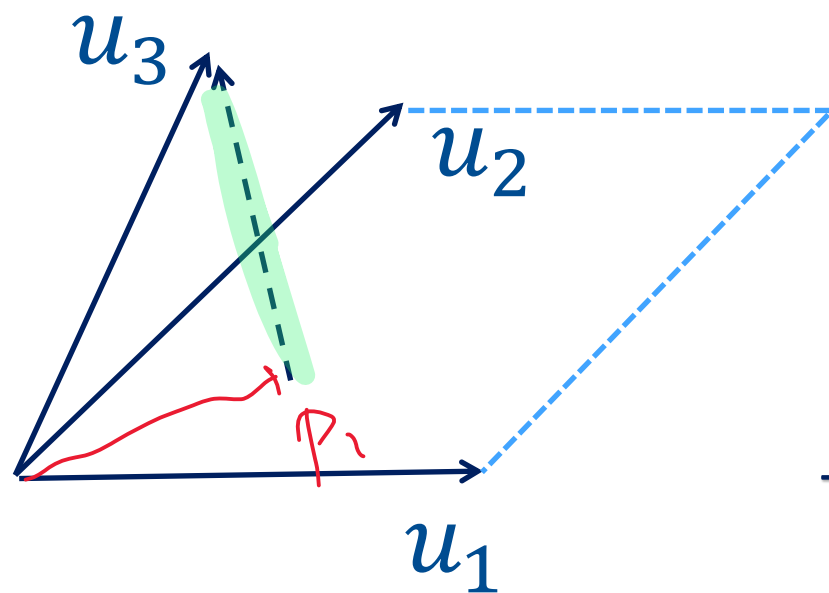
iii) Norm = 1.

Normalize to

$$\frac{u_2 - p_1}{\|u_2 - p_1\|}$$

k=3 Case: Derivation

Find orthonormal set U s.t. $\text{span}(U) = \text{span}(\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\})$.



$$v_1 = \frac{u_1}{\|u_1\|}$$

$$v_2 = \frac{u_2 - p_1}{\|u_2 - p_1\|}$$

$$v_3 = ?$$

$$\frac{u_3 - p_2}{\|u_3 - p_2\|}$$

To get $\underline{w}_3$ orthogonal
to v_1, v_2 : $u_3 - p_2$

new vector u_3 minus
its projection to old directions

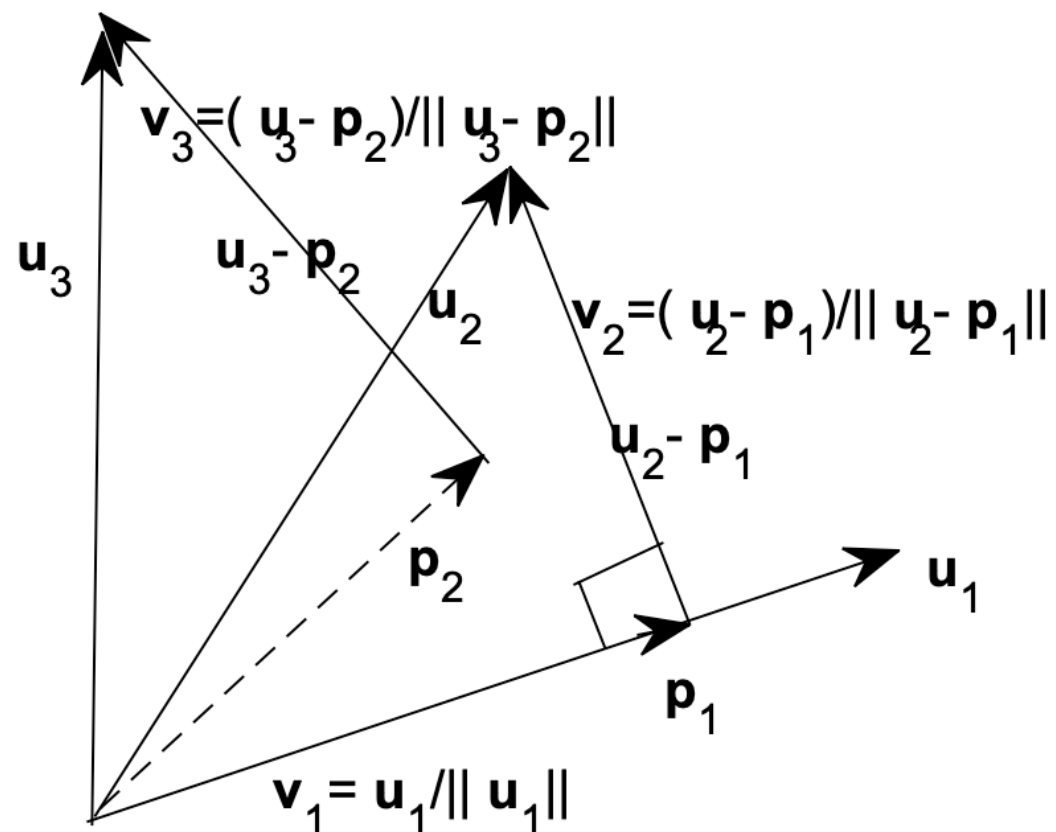
How to compute the projection to $\text{span}\{u_1, u_2\}$? (i.e. p_2)

Step 1: Find orthonormal basis of it, which is v_1, v_2

Step 2: Using an earlier formula, $p_2 = (u_3^T v_1) v_1 + (u_3^T v_2) v_2$.

k=3 Case: Result

Find orthonormal set U s.t. $\text{span}(U) = \text{span}(\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\})$.



$$v_1 = \frac{u_1}{\|u_1\|}$$

$$v_2 = \frac{u_2 - p_1}{\|u_2 - p_1\|}$$

where $p_1 = \text{Proj}_{\text{span}(u_1)}(u_2) = \langle u_2, v_1 \rangle v_1$

$$v_3 = \frac{u_3 - p_2}{\|u_3 - p_2\|}$$

new vector minus
projection to old directions

where $p_2 = \text{Proj}_{\text{span}(u_1, u_2)}(u_3) = \langle u_3, v_1 \rangle v_1 + \langle u_3, v_2 \rangle v_2$

Back to the Question: $k=3$ Case

Find orthonormal set V s.t. $\text{span}(V) = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

Steps:

Step 1: Find the orthonormal basis of $\text{span}\{\mathbf{u}_1\}$.

Step 2: Find the second vector in the ortho-basis of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$.

Step 3: Find the 3rd vector in the ortho-basis of $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$

This vector should be:

i) Orthogonal to $\mathbf{u}_1, \mathbf{u}_2$;

ii) in $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

iii) Norm = 1.

Normalize
to $\frac{u_3 - p_2}{\|u_3 - p_2\|}$

Answer: Direction same as $u_3 - p_2$
where $p_2 = \text{Proj}_{\text{span}(u_1, u_2)}(u_3)$
 $= \langle u_3, v_1 \rangle v_1 + \langle u_3, v_2 \rangle v_2$

new vector minus
projection to old directions

Exercise: k=4 Case

Find orthonormal set V s.t. $\text{span}(V) = \text{span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4)$.

Suppose we have found v_1, v_2, v_3 .

$$v_4 = \frac{u_4 - p_3}{\|u_4 - p_3\|}$$

where $p_3 = \sum_{i=1}^3 (u_4^T v_i) v_i$

or $p_3 = (u_4^T v_1) v_1 + (u_4^T v_2) v_2 + (u_4^T v_3) v_3$

Part I.2 Formal Description of G-S Process

Gram-Schmidt Process

Suppose $U = \{u_1, \dots, u_k\}$ is a set of linearly independent elements.

Question: Based on U , can we find an orthonormal set V such that $\text{span}(V) = \text{span}(U)$?

Gram-Schmidt Process

Input $U = \{u_1, \dots, u_k\}$

For $i = 1, \dots, k$

If $i = 1$:
 $p_0 = 0$

Else:

$$p_{i-1} = \sum_{j=1}^{i-1} \langle u_i, v_j \rangle v_j = \textit{Proj}_{\text{span}\{u_1, \dots, u_{i-1}\}}(u_i).$$

projection to old directions

$$\text{Set } v_i = \frac{u_i - p_{i-1}}{\|u_i - p_{i-1}\|} \in \text{span}(u_1, \dots, u_{i-1})^\perp$$

new vector minus projection to old directions

Return $V = \{v_1, \dots, v_k\}$

Gram-Schmidt Process

Suppose $U = \{u_1, \dots, u_k\}$ is a set of linearly independent vectors.

Question: Based on U , can we find an orthonormal set V such that $\text{span}(V) = \text{span}(U)$?

Proposition 17.3 (Gram-Schmidt Process)

The set $V = \{v_1, \dots, v_k\}$ returned by the Gram-Schmidt process is an orthonormal basis of $\text{span}(U)$.

Proof: Orthogonality: $u_i - \text{proj}_{\text{span}(u_1, \dots, u_{i-1})}(u_i) \perp \text{span}(u_1, \dots, u_{i-1})$,

their normalized versions are also orthogonal:

$$v_i = \frac{u_i - \text{proj}_{\text{span}\{u_1, \dots, u_{i-1}\}}(u_i)}{\|u_i - \text{proj}_{\text{span}\{u_1, \dots, u_{i-1}\}}(u_i)\|}. \quad v_i \perp v_1, \dots, v_{i-1}.$$

Norm: $\|v_i\| = 1$.

Basis: i) Independence. By orthogonality.

ii) Span. $\text{Span}(V) = \text{Span}(U)$. [skip proof]

Gram-Schmidt Process Example

Example Find the orthonormal basis of $\text{span}(S)$, where

$$S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^4$$

$\Leftrightarrow$ Find — of $C \begin{bmatrix} 1 & -1 & 4 \\ 1 & 4 & -2 \\ 1 & 4 & 2 \\ 1 & -1 & 0 \end{bmatrix}$

Gram-Schmidt Process Example

Example $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix} \right\} \subseteq \mathbb{R}^4$

In $\mathbb{R}^n$, the standard inner product is the scalar product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^T \mathbf{x}$.

Now find the orthonormal basis for $\text{Span}(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$.

Step 1: $\underline{p_0 = 0}, \quad \textcircled{v_1} = \frac{u_1 - p_0}{\|u_1 - p_0\|} = \frac{u_1}{\|u_1\|} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}.$

Step 2: $p_1 = \text{Proj}_{\text{span}(u_1)}(u_2) = \langle \underline{u_2}, v_1 \rangle v_1 = 3v_1.$

$$u_2 - p_1 = \begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix} - 3 \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} = \begin{bmatrix} -5/2 \\ 5/2 \\ 5/2 \\ -5/2 \end{bmatrix}, \quad v_2 = \frac{u_2 - p_1}{\|u_2 - p_1\|} = \frac{1}{5} \begin{bmatrix} -5/2 \\ 5/2 \\ 5/2 \\ -5/2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}.$$

Gram-Schmidt Process Example (cont'd)

Step 3: $p_2 = \text{Proj}_{\text{span}(u_1, u_2)}(u_3) = \langle u_3, v_1 \rangle v_1 + \langle u_3, v_2 \rangle v_2 = 2v_1 - 2v_2.$

$$u_3 - p_2 = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix} - \left(2 \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix} - 2 \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix},$$

$$v_3 = \frac{u_3 - p_2}{\|u_3 - p_2\|} = \frac{1}{4} \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix}.$$

$\therefore \left\{ \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \right\}$ is an orthonormal basis for $\text{span}(u_1, u_2, u_3)$.

Part I.3 Applications of G-S Process: Projection

Application of Orthonormal Basis: Compute Projection

Application question:

How to compute the projection of u onto a subspace $\text{span}(u_1, \dots, u_k)$?

Let $S = \{u_1, u_2, u_3\}$ be a set of vectors in $\mathbb{R}^4$, where $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix}$. Find

the projection of the vector $v = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 1 \end{bmatrix}$ onto $\text{span}(S)$.

Answer:

First, find an orthonormal basis $v_1, \dots, v_k$ by Gram-Schmidt process.

Second, use formula $p = \sum_{i=1}^k \langle u, v_i \rangle v_i$

Compute Projection onto a Span

Let $S = \{u_1, u_2, u_3\}$ be a set of vectors in $\mathbb{R}^4$, where $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 4 \\ 4 \\ -1 \end{bmatrix}$, $u_3 = \begin{bmatrix} 4 \\ -2 \\ 2 \\ 0 \end{bmatrix}$. Find

the projection of the vector $v = \begin{bmatrix} 3 \\ 2 \\ 1 \\ 1 \end{bmatrix}$ onto $\text{span}(S)$.

Answer: Since

$$\left\{ \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 1/2 \\ 1/2 \\ -1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \right\}$$

Use Gram-Schmidt process to find it

is an orthonormal basis for $\text{span}(u_1, u_2, u_3)$,

$$p = \langle v, q_1 \rangle q_1 + \langle v, q_2 \rangle q_2 + \langle v, q_3 \rangle q_3 = \frac{7}{2}q_1 - \frac{1}{2}q_2 + \frac{1}{2}q_3 = \begin{bmatrix} 9/4 \\ 5/4 \\ 7/4 \\ 7/4 \end{bmatrix}.$$

Framework To Summarize Orthonormal Basis

Inverse

Orthonormal Basis

Definition

Existence

N pivots,
Columns independent,
...

exists for
subspace of $\mathbb{R}^n$

Computation

- i) Product of E_i 's
- ii) GJ-E for $[A; I]$

Gram-Schmidt process

Application

Solution of $Ax=b$

- compute coordinate
- compute projection
- eigenvalue theory (later lectures)

Properties

$$D^{-1} = \dots$$
$$U^{-1} \ni \text{iff } \dots$$
$$(AB)^{-1} = B^{-1}A^{-1}$$

$$Q^T Q = I,$$
$$\|Qx\| = \|x\|, \dots$$

Part II Motivation of Determinant

2 by 2 Matrix: Invertible Condition

When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

First trial: by Def.

Step 1: Assume A^{-1} exists:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Step 2: Solve the system:

$$\begin{cases} ax + bz = 1, & cx + dz = 0 \\ ay + bw = 0, & cy + dw = 1 \end{cases}$$

~~Quite~~ complicated.

A bit

Use tools we learned!

2 by 2 Matrix: Invertible Condition

When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} \underline{a} & b \\ \underline{c} & d \end{bmatrix} \xrightarrow{R_2 - R_1 \cdot \frac{c}{a}} \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \end{bmatrix} \xrightarrow{R_2 \cdot a} \begin{bmatrix} a & b \\ 0 & ad - bc \end{bmatrix}$$

A invertible $\iff$ A has 2 pivots

$\iff$ columns are independent $\iff \dots$

Case 1: $a \neq 0$

$$A \longrightarrow \begin{bmatrix} a & b \\ 0 & ad - bc \end{bmatrix}$$

2 pivots $\iff ad - bc \neq 0$.

Case 2: $a = 0$

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} c & d \\ 0 & b \end{bmatrix}$$

2 pivots $\iff c \neq 0, b \neq 0 \iff ad - bc = -bc \neq 0$.

Conclusion: A invertible $\iff ad - bc \neq 0$.

Recall: Homework Problems

When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Answer: Iff $ad - bc \neq 0$.

Advantage over the 9 conditions for invertibility:
Closed-form expression to judge invertibility

Remark: We had an expression of inverse, but:
—no expression of “judging invertibility”;
—rely on an algorithm.

Motivation: Find an Invertibility Condition

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix.

Question 18.1: [extension of invertibility condition]

Can we extend $(ad - bc)$ to $n \times n$ matrix, i.e. find $\det(\cdot): \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$

(a function of a matrix) such that

$$\det(A) \neq 0 \text{ iff } A \text{ is invertible?}$$

Question 18.2: WHAT is $ad - bc$?

The fundamental understanding can help answer Q1.

How to Tackle Q2 at all?

Open-ended questions are often hard.

But there are patterns.

A powerful view to tackle such “what is” question:

GEOMETRY!

Understanding $ad - bc$

When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

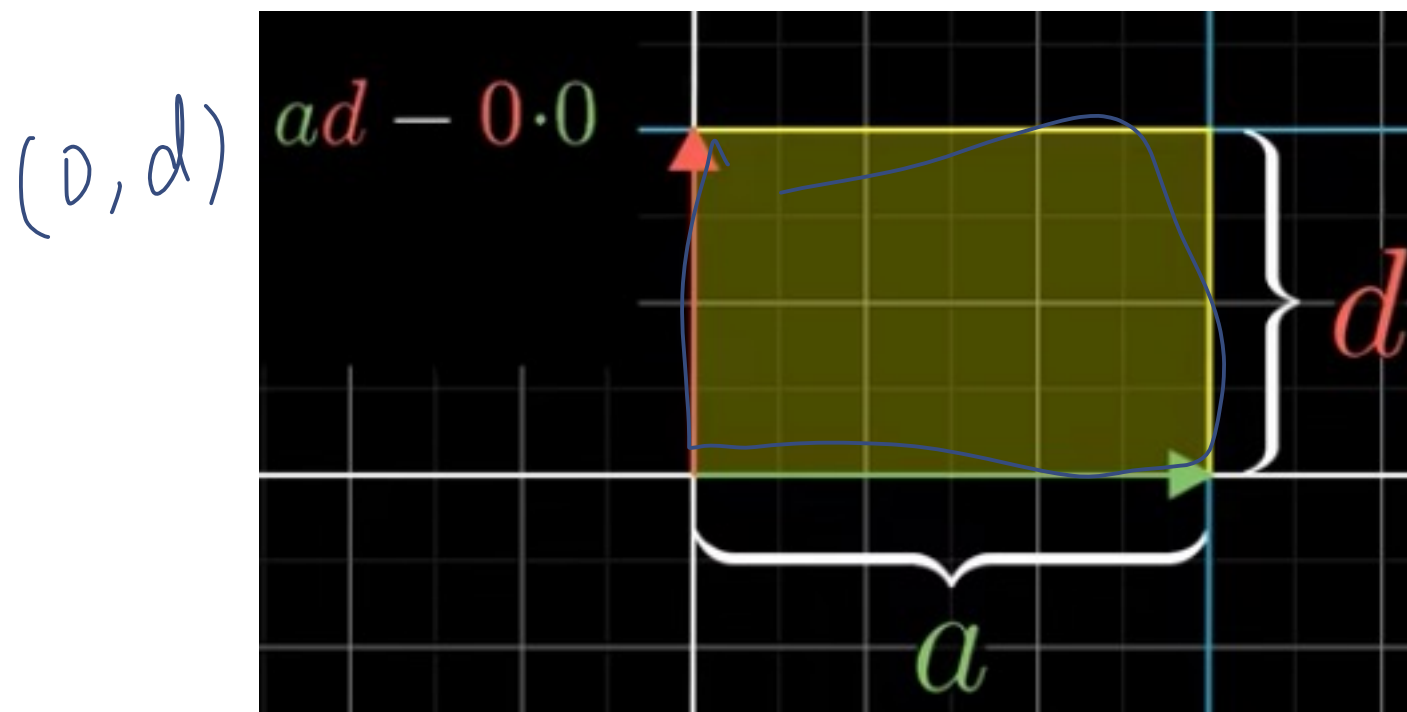
Answer: Iff $ad - bc \neq 0$

Question 18.2: WHAT is $ad - bc$?

Understand by GEOMETRY! (By Cauchy)

$$\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}^{-1} \quad ad \neq 0$$

Special case: $b = c = 0$



area of
rectangular is ad

$(a, 0)$

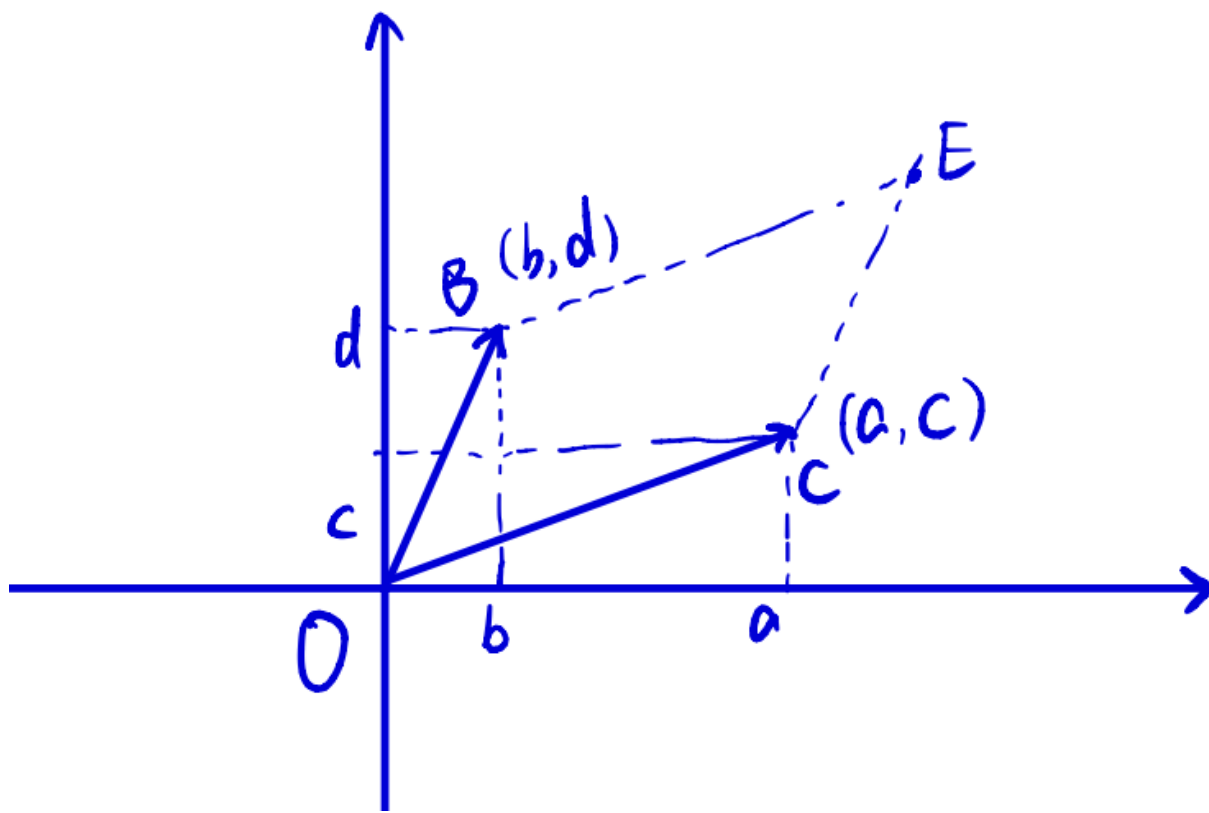
Understanding $ad - bc$

Homework problem: When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{Answer: Iff } ad - bc \neq 0.$$

Question 18.2: WHAT is $ad - bc$?

Understand by GEOMETRY! (By Cauchy)



Last page: rectangular. (if $b=c=0$).

For general a, b, c, d , column vectors $\begin{bmatrix} a \\ c \end{bmatrix}, \begin{bmatrix} b \\ d \end{bmatrix}$ form parallelogram $OCEB$

Exercise Try to find the relation of $ad - bc$ and parallelogram $OCEB$

Understanding $ad - bc$

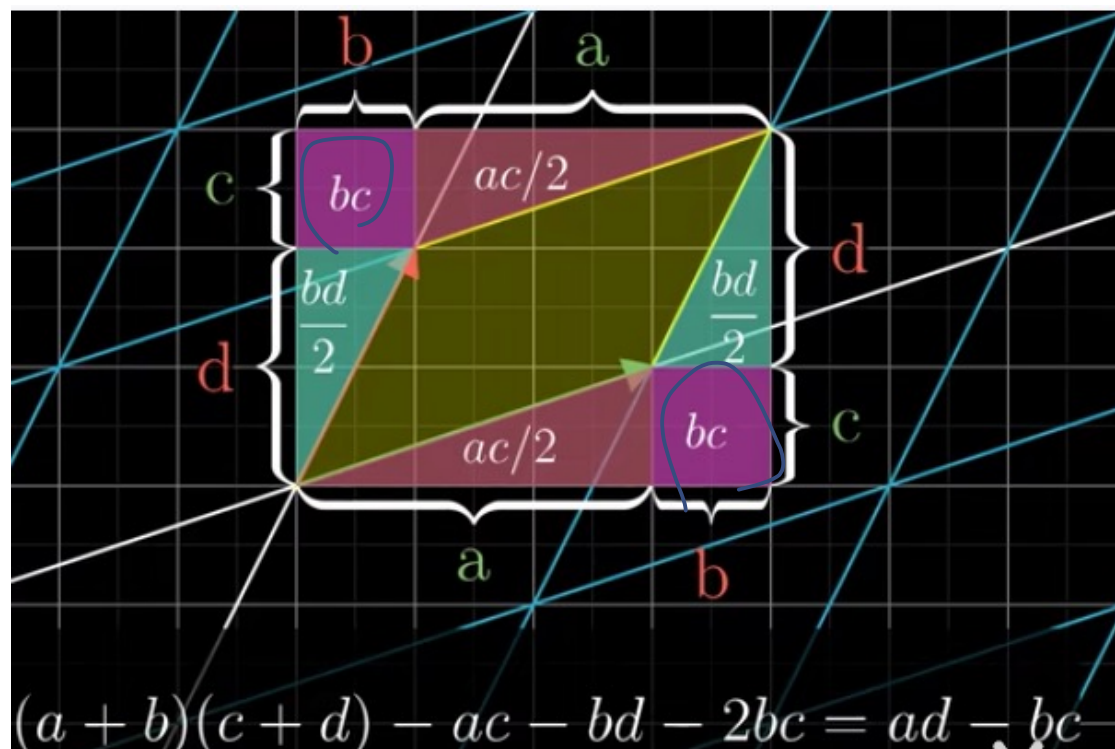
Homework problem: When is a 2×2 matrix invertible?

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{Answer: Iff } ad - bc \neq 0$$

Question 18.2: WHAT is $ad - bc$?

Understand by GEOMETRY! (By Cauchy)

Answer: The area of a parallelogram formed by the column vectors!



Claim

$ad - bc =$ Area of the parallelogram formed by column vectors $\begin{bmatrix} a \\ c \end{bmatrix}, \begin{bmatrix} b \\ d \end{bmatrix}$.

Large rectangle: $(a+b)(c+d) = \cancel{ac} + \underline{ad} + \cancel{bc} + \cancel{bd}$

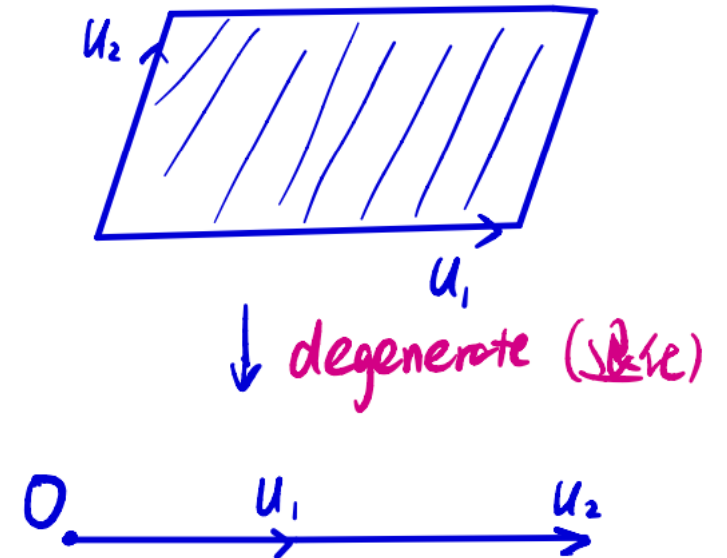
Other parts: $2bc + \underline{ac} + \underline{bd}$

Difference: $(a+b)(c+d) - (\cancel{2bc} + \cancel{ac} + \cancel{bd}) = \underline{ad} - \underline{bc}$

Degenerate $ad - bc$

Question 18.2: WHAT is $ad - bc$?

Area of the parallelogram
(formed by two column vectors from the matrix)!



What does $ad - bc = 0$ mean?

It means the area of the parallelogram is 0;

Equivalently, the two column vectors are parallel;
(dependent)

Equivalently, the matrix is NOT invertible.

Understanding $ad - bc$

Question 18.2: WHAT is $ad - bc$? Area of the parallelogram

When is a 2×2 matrix invertible? Iff $ad - bc \neq 0$.

New Interpretation: A is invertible iff the area of the parallelogram is non-zero.

Logic chain (逻辑链):

We observe: $ad - bc$ is the area of the parallelogram of columns.

$ad - bc = 0 \Leftrightarrow$ the area of the parallelogram of columns = 0.

$\Leftrightarrow$ columns $\begin{bmatrix} a \\ c \end{bmatrix}, \begin{bmatrix} b \\ d \end{bmatrix}$ are on the same line.

$\Leftrightarrow$ columns are dependent

$\Leftrightarrow A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is not invertible

We knew (from homework; algebraic proof)

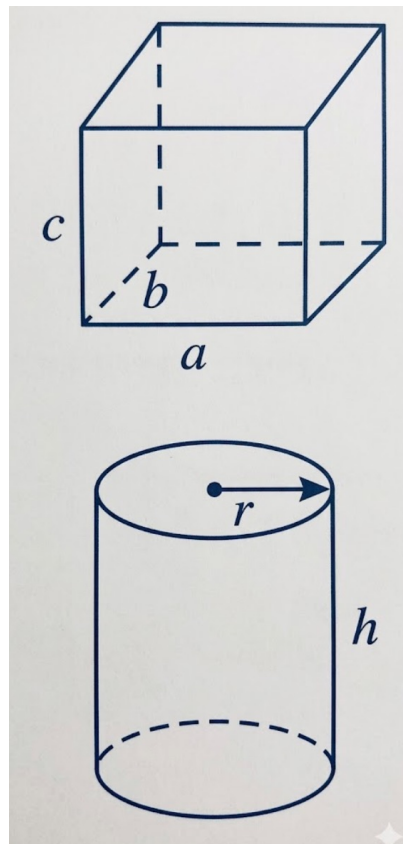
Extension to Higher Dimension

Question 18.1: [extension of invertibility condition]

Can we extend $(ad - bc)$ to $n \times n$ matrix, i.e. find function $\det()$ s.t.
 $\det(A) = 0$ iff A is invertible?

Answer: Let's extend the **geometrical meaning** of "area" to higher dimension!

Examples of computing **Volume** :



Cuboid (长方体)

$$V = abc$$

Cylinder (圆柱体)

$$\begin{aligned} V &= h \cdot (\text{Area of base disk}) \\ &= \pi r^2 h \end{aligned}$$

↓ extend

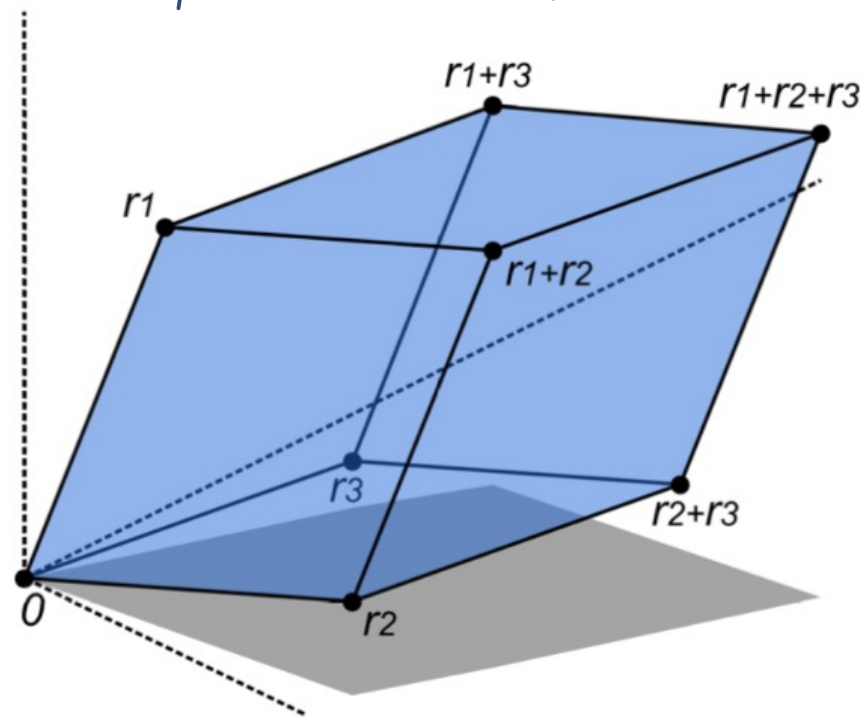
Volume

Extension to Higher Dimension

Question 18.1: [extension of invertibility condition]

Can we extend $(ad - bc)$ to $n \times n$ matrix, i.e. find function $\det()$ s.t.
 $\det(A) = 0$ iff A is invertible?

Extend parallelogram to polytope!



“Volume”!

Volume of polytope (多面体) formed by $\vec{r}_1, \vec{r}_2, \vec{r}_3$:

Next Time

What's the “volume” of the polytope formed by
 $a_1, a_2, \dots, a_n \in \mathbb{R}^n$?

This “volume” will be called “determinant” of
matrix $A=[a_1, a_2, \dots, a_n]$.

We will provide a formula for the determinant.

日期: /

Midterm exam

Sunday, Mar 29, 18-20:00.

Liwen Hall.

Closed book/notes.

Only pen & id.



Summary Today (Write Your Own)

One sentence summary:

Detailed summary:

Summary Today (Write Your Own)

One sentence summary:

We learned determinants.

Detailed summary:

Gram-Schmidt process.

- **Goal:** Construct orthonormal basis of a subspace, from a basis of the subspace.
- **Main trick:** every new vector minus projection to old directions
- **Theoretical result:** Gram-Schmidt process indeed returns an orthonormal basis.

Motivation:

- $ad - bc \neq 0$ is a simple criterion for invertibility.
- To extend it, we notice it indicates “area” of parallelogram.
- Motivating question: how to compute/define “volume” of polytope?